An Extension of Buddy Systems to Arbitrary Radix for Deterministic Memory Allocation

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1 Introduction

Unlike allocation models using variable-time search algorithms [1], this architecture employs a hierarchical occupancy map to resolve coordinates deterministically. Extending the traditional binary buddy system to an arbitrary radix replaces it with a geometric steering mechanism.

2 Architectural Axioms

These axioms establish the geometric and mathematical constraints of the lattice. They transform each memory request into a coordinate resolution task, avoiding search-and-sort operations to maintain an instruction determinism of exactly 13 steps.

2.1 Axiom I: The State Map (L) and Structural Arity ($\mathcal{A}$)

For modeling and testing purposes this paper demonstrates an n-level lattice L of fixed depth ($n = 3$) **defining** a total memory space. The architecture **maintains** a constant Arity ($\mathcal{A}$) of 32. A hierarchical progression of tiers **organizes** the lattice: the atomic Sector (S_0), the Row of Sectors (S_1), the Plane of Rows (S_2), and the Cube of Planes (S_3). Lattice coordinates (p, r, s) **determine** the address of any specific leaf node, where $p, r, s \in \{0, \ldots, \mathcal{A} - 1\}$. The geometry **maintains** invariance regardless of the physical byte-width the sector **occupies**.[†]

2.2 Axiom II: Hierarchical Occupancy Counters (Γ)

Discrete occupancy counts (Γ) at every tier **enable** constant-time steering. Following bit-block representation principles [2], these counters **generate** a real-time density map of the lattice:

- **Level 1 (Row):** $\Gamma_1 = \sum_{i=0}^{\mathcal{A}-1} \Gamma_{0,i}$
- **Level 2 (Plane):** $\Gamma_2 = \sum_{j=0}^{\mathcal{A}-1} \Gamma_{1,j}$
- **Level 3 (Cube):** $\Gamma_3 = \sum_{k=0}^{\mathcal{A}-1} \Gamma_{2,k}$

This hierarchical summation lets the Global Registrar to **execute** an admissibility check in a single instruction cycle, **fulfilling** the fast-fail condition established in the audit.

[†]The Large allocator functions as a cross-tier searcher for sequence resolution, managing the fractal logic across the Plane manifold to satistfy multi-Row requests.

2.3 Axiom III: Discrete Enforcement of the Resilience Floor (Φ)

The Iscandar system **enforces** a mandatory occupancy boundary, the Resilience Floor (Φ). Traditional allocators search for gaps as memory nears capacity; in contrast, this system **triggers** a deterministic refusal once Γ_3 reaches the Φ threshold.

2.4 Axiom IV: Rotational Steering ($\vec{R}$)

A rotational steering vector **balances** the lattice structure, preventing localized occupancy "hot spots" and maintaining the LHL/RHL Handshake Oracle within its optimal operational tax. The "tax" (instruction count) depends on the Handshake Oracle resolving a valid "weld" or "gap" within a single word-width scan of magnitude $\mathcal{A}$. If allocations were to "pile up" linearly at the beginning of every Plane, high-density clusters would emerge, necessitating bit-shuffling or metadata updates.

2.5 Axiom V: The Contiguity Guarantee

Any resolved coordinate sequence **represents** a continuous physical block in the Iscandar lattice. The Level 1 LHL/RHL Handshake Oracle **enforces** this property, **validating** that the combined leading and trailing fragments across a sector-join **equal** the requested magnitude M without gaps.

3 Inductive Proof of Deterministic Resolution

Theorem: A fixed-depth $\mathcal{L}$ of Arity $\mathcal{A} = 32$ **establishes** deterministic resolution for operator Ω.

3.1 The Complexity Proof

The $\mathcal{L}$ **maps** its branching factor to the processor word-width to **optimize** the classical buddy system search [3]. Let $\mathcal{L}_n$ **designate** a sub-lattice of level n, where resolution time $T(\mathcal{L}_n)$ **represents** the total instruction count (ι) required to **resolve** a memory coordinate. Resolution time $T(\mathcal{L}_n)$ **represents** the instruction count (ι) for **resolving** a memory coordinate in a level-n sub-lattice.

1. **Base Case ($n = 1$):** At the Row level ($\mathcal{L}_1$), a hardware bit-scan primitive **resolves** a free sector; this operation **maintains** atomicity on 32-bit architectures.

$$T(\mathcal{L}_1) = I_{base} \tag{1}$$

2. **Inductive Step:** Given the invariant that resolution time for a level-k structure is $T(\mathcal{L}_k) = kI_{base}$, resolving a request at level $k + 1$ (e.g., the Plane level) **requires**:

   - The occupancy sum Γ_{k+1} is **evaluated** via bitmask comparison to **select** a viable Level-k child. When $\mathcal{A}$ **aligns** with a power of 2, this lookup and steering operation **consumes** a single table lookup [4].
   - The resolution operation **descends** to the level-k $\mathcal{L}$ identified by the steering vector from the Oracle.
   - The resulting recurrence relation is:

$$T(\mathcal{L}_{k+1}) = I_{base} + T(\mathcal{L}_k) = I_{base} + kI_{base} = (k + 1)I_{base} \tag{2}$$

   - **Termination Constraint:** The recursion **terminates** at $\mathcal{L}_1$, where the hardware bit-scan **yields** the final effective address offset.

3.2 Conclusion of Determinism

By induction, the resolution time for any n-level $\mathcal{L}$ structure is strictly nI_{base}. While standard stochastic allocators exhibit execution times as a function of free-block count (N)—typically yielding $O(\log N)$ or worse in fragmented states [5]—the Iscandar architecture **collapses** this complexity. n as a constant **enforces** deterministic latency of resolution regardless of heap saturation.

In the Iscandar model, n is an architectural constant (e.g., $n = 3$); so total execution time is:

$$T_{total} = 3I_{base} \tag{3}$$

This result confirms that for a Cube of $\mathcal{A}^k$ Sectors, the time required to resolve an address is invariant. This performance is maintained provided the system stays in the operational domain of the Resilience Floor (Φ). By trading spatial capacity for temporal certainty, the architecture **eliminates** the "temporal jitter" that characterizes non-deterministic real-time systems.

3.3 Temporal Consistency and Boundary Management

To **preclude** latency variability during boundary-straddling, Iscandar **utilizes** a computational handshake or "Oracle," a precomputed resolution-path for non-aligned straddle cases, **ensuring** the instruction count for a $\mathcal{A}$-sector block remains **isochronous** regardless of its alignment relative to the Tier-1 registers. By **eliminating** the alignment-check penalty, the architecture **preserves** a deterministic execution window across the entire addressable space.

Here we describe the "Factory" math used to pre-calculate the Oracle tables, transforming a search problem into a static retrieval of pre-verified geometric facts.

3.3.1 Generating the Boundary Tables (RHL/LHL)

The boundary tables are 1D vectors that map every possible bit-permutation of a Guard Shard to its boundary potential.

- **Right-Hand Lookahead (RHL):** For any shard state S, the table stores the starting index of the longest contiguous zero-run terminating at the right-hand boundary. If the terminal bit is **1**, the value is **Null**.
- **Left-Hand Lookahead (LHL):** Maps the shard state to the final bit position of the zero-run starting from the left-hand boundary. If the initial bit is **1**, the value is **Null**.
- **Math:** These are pre-calculated *Suffix/Prefix Automata* that eliminate runtime scanning [**?**].

3.3.2 Generating the Core Oracle (2D Matrix)

The Core Shard utilizes a 2D matrix ($2^{\mathcal{A}/4} \times \mathcal{A}/4$) to resolve internal gaps.

- **Mechanism:** Exhaustive enumeration of all $2^{\mathcal{A}/4}$ bit-patterns against all possible length requests within the core.
- **Math:** Each cell (i, j) is filled using a windowed zero-run algorithm. This transforms the shard into a lookup-based *Finite State Transducer* [**?**].

3.3.3 The Cube Case (The Boolean Primitive)

The "Cube Case" represents the **Ultimate Degeneracy**. It is the boolean truth for the entire $\mathcal{A}^k$ bitspace.

- **Scale:** A single bit of metadata representing the occupancy of the entire $\mathcal{A}^k$ volume.
- **Boolean Law:**
  - **0 (Open):** The entire volume is zero.
  - **1 (Closed):** Any bit within the volume is one.
- **Math:** This represents the collapse of a high-dimensional state space into a *Degenerate Polytope*, providing the structural filler for Huge Case construction with $O(1)$ verification [**?**].

4 Deterministic Coalescence and the Resilience Floor

Maintaining temporal determinism in a dynamic environment **requires** a deallocation process (coalescence) **mirroring** the efficiency of the allocation logic. Operational stability of the iscandar lattice uses discrete enforcement of the Resilience Floor (Φ), **precluding** the high-entropy states typically associated with stochastic fragmentation [5].

4.1 Symmetric Coalescence Mechanics

In the iscandar architecture, coalescence is not a variable-time search for adjacent free blocks, but a **deterministic state update**. Whether the Oracle **descends** to clear a bit-position or **propagates** the release upward through the hierarchy, the operation **remains isochronous**. Because the bit-address of any Sector (S_0) is a fixed function of its coordinates, the update to the occupancy counters (Γ) **consumes** the same instruction count (I_{base}) as the initial resolution.

1. The atomic leaf node state Γ_0 is **toggled** to 0 (Available).
2. The change **propagates** immediately to the parent Row counter (Γ_1).
3. The Level-2 (Plane) and Level-3 (Cube) counters are **updated** through integer subtraction.

Because the lattice depth is fixed at $n = 3$, time complexity for allocation are identical to deallocation ($3I_{base}$). This symmetry **ensures** that system overhead remains constant regardless of memory "churn," **eliminating** the temporal spikes typically made by coalescing algorithms [6].

4.2 Operational Enforcement of Ω

The Resilience Floor (Φ) **serves** as the authoritative boundary of the system's operational domain. While traditional allocators attempt to satisfy requests until the address space is 100% utilized—leading to $O(N)$ search patterns as fragmentation increases [7]—the iscandar lattice **utilizes** the Γ_3 counter as a deterministic "circuit breaker."

The decision logic for the Resolution Operator (Ω) is:

$$\Omega(\text{request}) = \begin{cases} \text{Resolve}(p, r, s) & \text{if } \Gamma_3 < K \\ \text{Refuse}(\text{Constant Time}) & \text{if } \Gamma_3 \geq K \end{cases} \tag{4}$$

where K is the sector threshold at which the Resilience Floor is triggered ($K = \mathcal{A}^k \times (1 - \Phi)$)

4.3 Summary of Empirical Performance

This enforcement mechanism ensures the system never enters the "fragmentation trap." By refusing to allocate beyond the floor (Φ), **iscandar** guarantees that the internal density of the $\mathcal{A}$-way branching structures never reaches a threshold where bit-mask steering degrades to a linear probe.

This confirms a **flat temporal profile** until the moment of refusal.

Proof of Coalescence Invariant

Let Γ_{total} be the sum of all leaf nodes $S_{0,i}$. At any discrete time t, the hierarchical counters **must** satisfy the invariant:

$$\Gamma_{Apex} \text{ (a.k.a. } \Gamma_3) = \sum \Gamma_2 = \sum \Gamma_1 = \sum \Gamma_0 \tag{5}$$

Since deallocation is defined as:

$$\Gamma_x(t+1) = \Gamma_x(t) - 1 \quad \text{for all levels } x \in \{0, 1, 2, 3\}$$

along the specific resolution path, the invariant is **preserved**. Because this path is fixed at depth n, the lattice is restored to a consistent, deterministic state in nI_{base} time.

5 Observations

Earlier sections established Iscandar as deterministic within the primary scope of this paper. However, it exhibits properties warranting distinct architectural observations.

5.1 As an Extension to Binary Buddy Systems

The binary buddy system, pioneered by Knowlton [8] and Knuth [9], sought to minimize search latency through a power-of-two partitioning scheme ($R = 2$). While this approach facilitates efficient coalescence, its search complexity remains fundamentally constrained to $O(\log_2 n)$. As modern memory address spaces scale alongside processor word sizes, the increasing logarithmic depth of these trees drives significant latency. Furthermore, at high utilization, binary buddy systems trigger "stalling" behaviors; the search algorithm must traverse increasingly fragmented leaf nodes, which inevitably erodes deterministic performance.

We generalize buddy system principles to an **Arbitrary Radix** ($R = \mathcal{A}$). By shifting from a binary tree to a high-radix fractal lattice, we collapse the depth of the search space to a fixed architectural constant n. This structural shift moves memory management from the domain of probabilistic searching to the domain of **Geometric Coordinate Resolution**.

5.2 As a Symmetric Voronoi Partition

As detailed in the appendices, this lattice functions as a **Symmetric Voronoi Partition**, where the adjacency of "buddies" is a pre-baked geometric property of the address space rather than a calculated proximity. In this model, adjacency is defined by a partition's total volume **associating** with its handle (a wrapped pointer), rather than the Euclidean distance of individual points from a centroid (e.g., the proximity of residences to a central post office).

5.3 As Linear Programming

The operational stability of the system is maintained via an **Interior Point Method**—analogous to projective transformations used in Karmarkar's algorithm [10]. By enforcing a **Resilience Floor** (Φ), we **ensure** the resolution path avoids "edges" of the feasible region (the fragmentation trap). Here, the allocator does not merely manage space; it **navigates** a high-dimensional manifold where the floor serves as a protective barrier against non-deterministic search behaviors.

5.4 Amortized Performance: Constant Time

The Iscandar architecture utilizes a *Seamless Sack* model that achieves amortized constant time for memory operations through a Next-Fit Circular Cursor mechanism.

- **Mechanism:** The cursor maintains forward momentum across the $\mathcal{A}^k$-sector manifold. By never re-scanning fragmented entropy within a single "lap," the system distributes the metabolic cost of search across its entire operational lifecycle.
- **Forward-Pressure Principle:** The system treats fragmentation as a transient spatial cost. It offsets the latency of filtering fragmented sectors by performing high-efficiency, contiguous allocations in reclaimed or vacant regions.

5.5 Performance Bound: Sub-Linear Determinism

The architecture maintains determinism through hierarchical skip-logic at intermediate levels. This mechanism constrains the worst-case search complexity to the k-th root of the total lattice volume, where k represents the depth of the fractal hierarchy.

- **Mechanism: Dimensional Collapse.** The architecture maps the 1D address space to a k-dimensional manifold ($\mathcal{A}^k$). Metadata sentinels at the Plane and Row levels enable the search engine to bypass saturated volumes without bit-level interrogation.
- **Logic for Cubic Architecture:**
  1. **Plane-Skip:** Evaluates $\mathcal{A}^2$-sector manifolds; the engine executes an $O(1)$ leap if free space is less than the request size.
  2. **Row-Skip:** Evaluates $\mathcal{A}$-sector rows; the engine executes an $O(1)$ leap if row density proves prohibitive.
  3. **Bit-Scan:** The engine restricts final verification to a single bit-vector of magnitude $\mathcal{A}$.
- **Deterministic Ceiling:** For a lattice of volume $N = \mathcal{A}^k$, the architecture caps the maximum investigation path at $k \times \mathcal{A}$, limiting the worst-case cost to $k \times \mathcal{A}$ operations.

5.6 As Supporting Six Sigma Design

As detailed in the appendices, the compactness of the iscandar kernel enables **complete path coverage testing** that operates in tandem with the previously established formal methods. The convergence of exhaustive path verification and formal proofs provides the necessary rigor to certify the algorithm as a **fault-tolerant** memory system, suitable for high-reliability applications.

5.7 5.7 Scalability via Depth Extension

The **iscandar** architecture scales by increasing the lattice depth up to two additional levels. Because the resolution time is a linear function of depth ($k \times I_{base}$), extending the lattice to $k = 4$ or $k = 5$ provides a massive increase in addressable memory volume while maintaining a strictly deterministic and predictable execution profile.

5.8 As Supporting Parallel Execution

The architecture allows for the distribution of the management load through parallel instantiation. Lesser structures can be managed by hardware subroutines to distribute the processing requirements across multiple units. These parallel hardware components can also supervise local memory independently, communicating management data upward through the hierarchy only as required.

6 Conclusion

The **iscandar** architecture demonstrates that the non-deterministic latency inherent in traditional memory management is a byproduct of stochastic search methodologies rather than a physical necessity. Replacing variable-depth search algorithms with a fixed-depth, $\mathcal{A}$-radix fractal lattice establishes deterministic execution.

7 Glossary of Terms

Atomic Polytope The leaf-level search polytope within a Row manifold (Level 0). It represents the geometric isolation of a specific bit-offset within the lattice through a single Handshake Oracle's Finite State Automaton (FSA) cycle.

Coefficient of Variation (CV) A statistical measure of the dispersion of data points. In the Iscandar context, a CV of zero confirms that the instruction path length is identical across all execution cycles, proving absolute determinism.

Contiguity Guarantee The formal assurance (Section 2.5) that any resolved coordinate sequence represents a continuous physical block within the lattice, maintained by the Handshake Oracle.

Fixed Instruction Tax ($K = 13$) The invariant number of macro-instructions required to resolve any memory request, regardless of request size or the current memory occupancy.

Global Registrar The Level 4 (Cube Scale) apex of the hierarchy. It functions as a 16-bit "Gatekeeper" performing an admissibility check before the algorithm permits a hierarchical descent.

Handshake Oracle (LHL/RHL/Center) A Finite State Automaton, being a precomputed body of solutions used at the Tiny (Level 0) and Small (Level 1) allocator levels to bridge "fractured" sequences between sectors. It utilizes Left-Hand Lookahead (LHL), Right-Hand Lookahead (RHL) and Center Lookahead constants to weld disparate fragments into a contiguous magnitude. In microcontroller implementations these are laid out as ROM tables.

Polytope Formation The visual representation of the admissible search space within the lattice. Each hierarchical tier (Cube, Plane, Row) can manifest polytopes that collapse as memory occupancy increases.

Resilience Floor (Φ) The mandatory occupancy boundary (Section 4.2) that prevents the system from entering a high-entropy state. Once the floor is reached, the system issues a deterministic refusal rather than engaging in variable-time search.

Uranus 5 The laboratory verification suite written in APL used to empirically show deterministic performance in the Iscandar architecture.

Appendix A: The 13-Instruction Mechanism

The following listings represent the raw executable logic of the iscandar mechanism. Each line corresponds to a discrete macro-instruction (K).

```
∇    Coord ← ALLOC_ROTATING_COUNTED
[1]  S3 ← (32 | SEED_P)          % K1: Resolve Tier-3 Plane index
[2]  S2 ← (32 | SEED_R)          % K2: Resolve Tier-2 Row index
[3]  S1 ← 1ι ∼Lattice[S3;S2;]    % K3: Resolve Tier-1 Bit index
[4]  Coord ← (S3, S2, S1)        % K4: Assemble 3D Coordinate
[5]  Lattice[S3;S2;S1] ← 1       % K5: Occupy sector
[6]  T2[S3;S2] -← 1              % K6: Update Tier-2 counter
[7]  T3[S3] -← 1                 % K7: Update Tier-3 counter
∇

∇    FREE_ROTATING_COUNTED Coord
[8]  (S3 S2 S1) ← Coord          % K8: Decompose Coordinate
[9]  Lattice[S3;S2;S1] ← 0       % K9: Release sector
[10] T2[S3;S2] +← 1              % K10: Increment Tier-2 counter
[11] T3[S3] +← 1                 % K11: Increment Tier-3 counter
[12] SEED_P ← (32 | S3 + 1)      % K12: Advance T3 seed
[13] SEED_R ← (32 | S2 + 1)      % K13: Advance T2 seed
∇
```

Appendix B: Updating Memory State Spaces by Mating the Oracle Tables

This section describes how **iscandar** resolves memory requests by intersecting partitioned subspaces. The system utilizes primary ratios relative to the structural Arity ($\mathcal{A}$): the Guard Shards ($0.5\mathcal{A}$) and the Core Shard ($0.25\mathcal{A}$).

Polytope Formation at Cube Level

The Cube Scale Allocator (Global Registrar)

Domain: Total Cube Volume resolution (addressable at $\mathcal{A}^k$).

Summation Predicate:

$$|C| = \sum_{i=0}^{\mathcal{A}-1} (\text{Sector Count of Plane } i)$$

Discussion: The Cube Scale Allocator computes the unique **"Gatekeeper"** sector count of the architecture. It provides the primary fast-fail heuristic for the system; if a request (S, M) is determined to be non-admissible at this tier, the algorithm immediately terminates without initiating a descent. This makes the registrar the most efficient filter in the hierarchy, ensuring that the cost of coordinate resolution is only paid when success is mathematically guaranteed.

Polytope Formation At Row and Plane Levels

The Large Resolution ($\mathcal{A} < L \leq 2\mathcal{A}$)

Domain: Row-join and sequence resolution within an $\mathcal{A}^2$-sector Plane manifold.

Summation Predicate:

$$C_{plane} = \sum_{i=0}^{\mathcal{A}-1} (C_{row,i} \cdot [y_i = 1])$$

Discussion:

- **Mechanism:** The *Structural Bridge*.
- **Geometric Event:** *Boundary Handshake* between adjacent vectors.
- **Resolution:** The system constructs the span via $P + T$. The RHL of Vector A (Preamble) mates with the LHL of Vector B (Tail) to verify connectivity across the boundary [**?**].

This tier represents a fractal expansion of the Level 1 logic. It treats individual rows as the atomic units of resolution, welding row-sequences to satisfy multi-row allocation requests. The Large allocator functions as a resolution engine, calculating a dynamic subset of Rows within the Plane to satisfy continuity requirements.

The Huge Resolution ($L > 2\mathcal{A}$)

Domain: Plane-join and sequence resolution within an $\mathcal{A}^k$ Cube manifold.

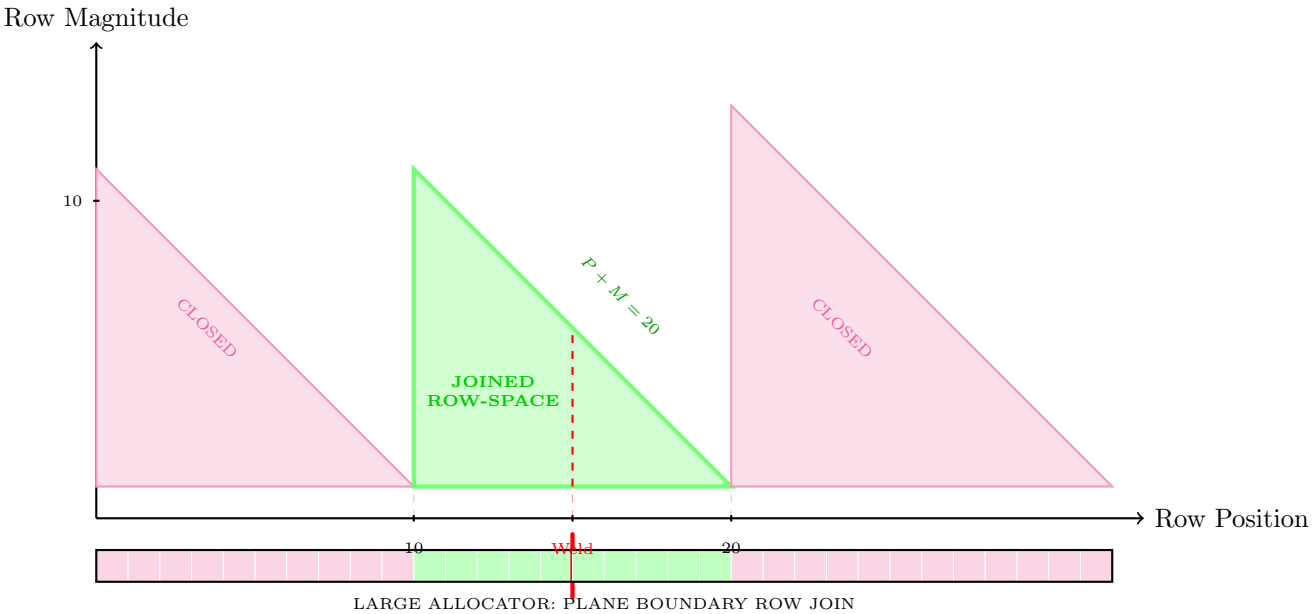
Summation Predicate:

$$C_{huge} = \sum_{i=0}^{\mathcal{A}-1} (C_{plane,i} \cdot [z_i = 1])$$

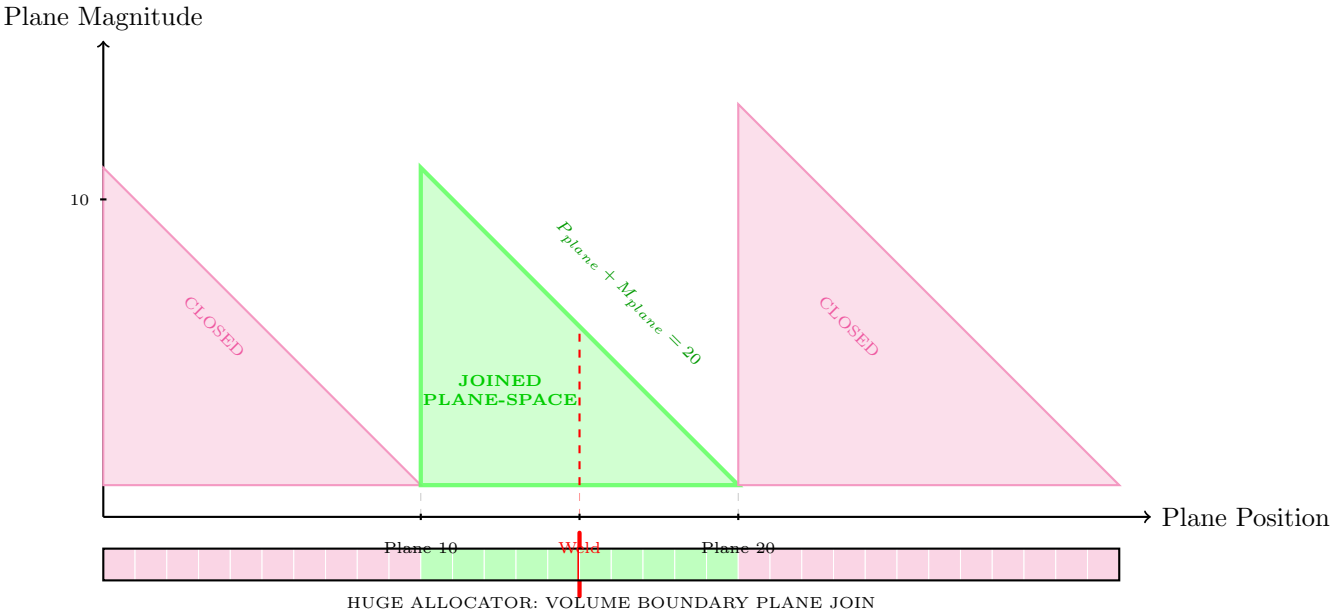
Discussion:

- **Mechanism: The Recursive Span**
- **Geometric Event:** *Multi-Vector Extrusion*
- **Resolution:** The system constructs the span via $P + (m \times \mathcal{A}) + T$. The Oracle verifies the m intervening words as "Clear Rows" ($\mathcal{A}$ nullity checks), treating the span as a single geometric object via induction [**?**].

This represents the final join-level resolution of the hierarchy. It manages the 32-radix plane counts, welding 2D plane segments into a unified 3D volume request. By treating planes as the fundamental unit of resolution at this tier, the system ensures that even the largest possible allocations are resolved with identical determinism. The Huge allocator calculates the specific number of Planes required to weld a unified 3D volume, treating each Plane count as a subordinate metric for the total Cube count.



LARGE ALLOCATOR: PLANE BOUNDARY ROW JOIN



HUGE ALLOCATOR: VOLUME BOUNDARY PLANE JOIN

Polytope Formation Between Rows

The Small Resolution ($L \leq \mathcal{A}$)

Domain: Sector-join and sequence resolution within an $\mathcal{A}$-sector Row manifold.

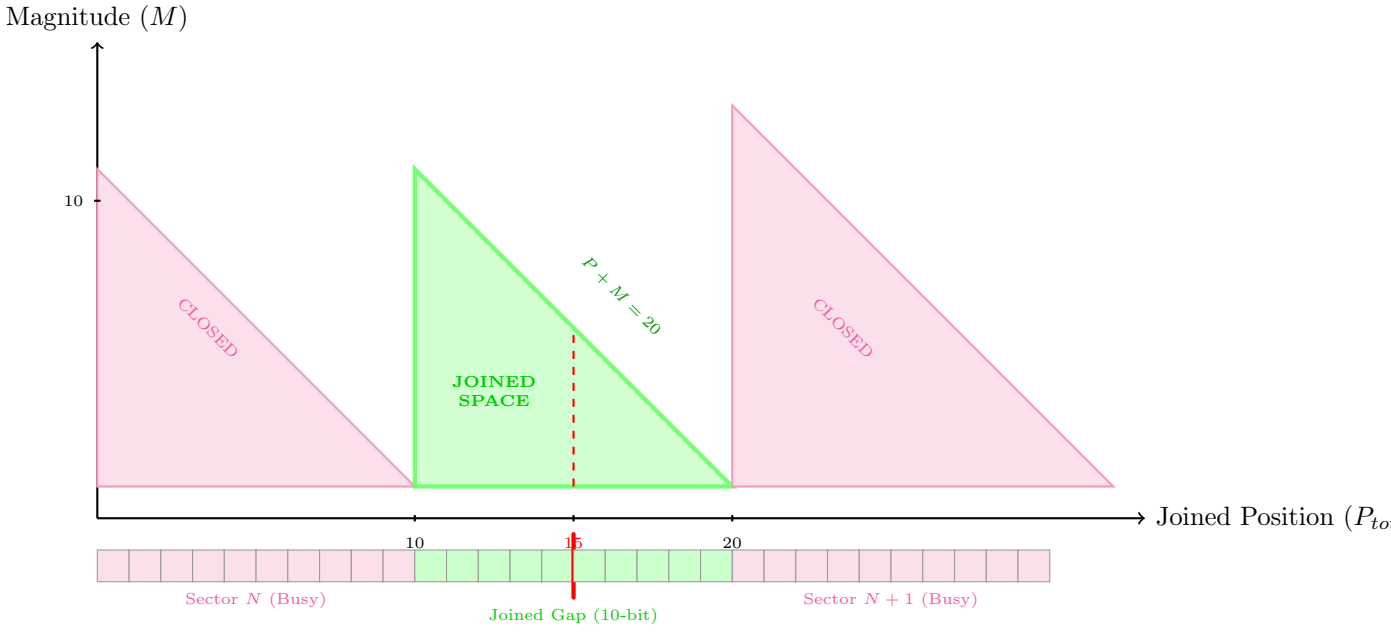
Summation Predicate:

$$C_{row} = \sum_{j=0}^{\mathcal{A}-1} (C_{bits,j} \cdot [x_j = 1])$$

Discussion:

- **Mechanism:** The *Trinitarian Handshake*.
- **Geometric Event:** A *Hyperplane Intersection*.
- **Resolution:** The Oracle intersects the Right-Hand Lookahead (RHL) of the first Guard, the internal gaps of the Core, and the Left-Hand Lookahead (LHL) of the second Guard. This operation intersects three half-spaces to verify a contiguous span [**?**].

This tier resolves the "fracture" typically found at sector boundaries. It functions as the primary submachine for the first level of welding, utilizing the **LHL/RHL Handshake Oracle** to join partial sector tails and heads into a single contiguous magnitude. By treating the row as a unified geometric manifold, the system ensures that boundary seams are transparent to the resolution logic.



Polytope Formation Within Row

The Tiny Resolution ($L \leq 0.25\mathcal{A}$)

Domain: The bit-lattice contained within a single Row manifold.

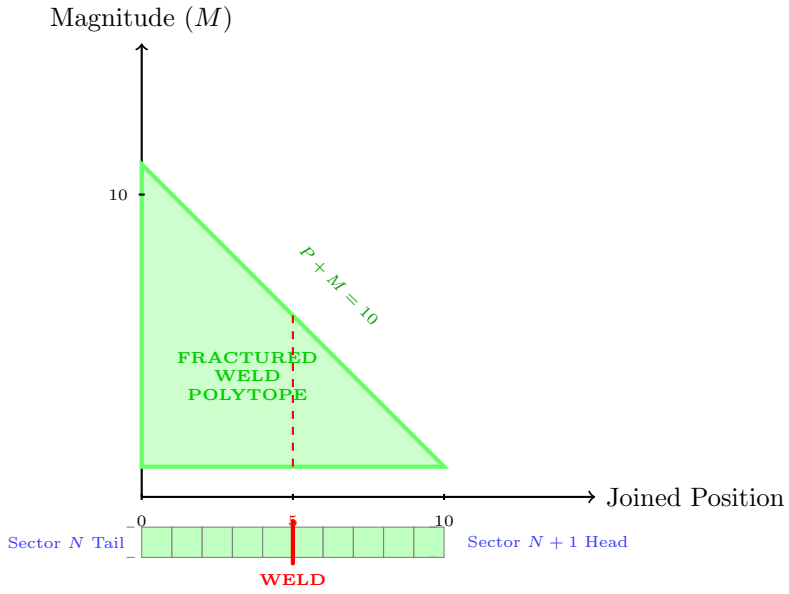
Summation Predicate:

$$C_{bits} = \sum_{i=0}^{\mathcal{A}-1} (\neg b_i \cdot [w_i = 1])$$

Discussion:

- **Mechanism:** The request falls entirely within the Core Shard (S_c).
- **Geometric Event:** An *Interior Locality* query.
- **Resolution:** The 2D Oracle Table functions as a point-in-polytope calculator. The system requires only the center lookup to return the $(position, length)$ tuple [**?**].

This represents the leaf-level execution of the deterministic descent. The **Center Handshake Oracle** identifies contiguous zero-bit sequences within a sector. This forms the Atomic Polytope of a search, resolving the physical bit-offset in a fixed hardware cycle.



Appendix C: The $K = 13$ Detailed Description of Instructions

To perform an "honest" audit for the monograph, we must map each of the 13 macro-instructions ($K = 13$) to the specific discrete predicates of the iscandar hierarchy established in the Architectural Axioms. This ensures the hardware execution is a transparent reflection of the underlying mathematics. The audit assumes a 32-radix descent where each level acts as a "Gate" that must be cleared to reach the subsequent tier.

7.1 Phase I: The Admissibility Gate (Level 3)

Predicate: $M \in \Gamma_{Apex} \iff \Gamma_{Apex} \leftarrow \Gamma_{Apex} + M$

1. **Instruction 1:** Load Global Registrar: Fetch the 16-bit register located at the Apex sector address as defined in the Cube Scale Allocator.
2. **Instruction 2:** Admissibility Comparison: Compare against the requested magnitude M. If False: Immediate Exit (Fast-Fail) as required by the Global Registrar's heuristics. If True: Increment by the requested magnitude.

7.2 Phase II: The Plane-Level Resolution (Level 2)

Predicate: $P_{Plane} \iff \sum_{i \in Plane}(Row[Inclusion]) \geq M$

3. **Instruction 3:** Index Plane Metadata: Using the address bits, index into the 32-radix plane metadata table.
4. **Instruction 4:** Load Row-Summary: Fetch the 15-bit metadata for the target plane summarizing the 32 constituent rows as established in Axiom II.
5. **Instruction 5:** Candidate Selection: Evaluate the plane-state to identify the first row-sequence that satisfies the weld.
6. **Instruction 6:** Heuristic Branch/Scan: Evaluate the 32-radix plane metadata scan to confirm the steering vector.

7.3 Phase III: The Row-Level Weld (Level 1)

Predicate: $Weld \iff RHL(Row_n) + Intermediates + LHL(Row_{n+m}) \geq M$

7. **Instruction 7:** Leading Fragment Access (RHL): Perform a Look-Up Table (LUT) lookup on the starting row's tail.
8. **Instruction 8:** Trailing Fragment Access (LHL): Perform a LUT lookup on the final row's head.
9. **Instruction 9:** Scalar Summation: Add Lead + Intermediates + Trail, maintaining the Contiguity Guarantee.
10. **Instruction 10:** Weld Validation: Compare the result against M to finalize the inter-row continuity.

7.4 Phase IV: The Bit-Lattice Isolation (Level 0)

Predicate: $P_{Tiny} \iff \sum_{i=start}^{end} \neg b_i = M$

11. **Instruction 11:** Center LUT Index: Access the bit-occupancy table for the target sector.
12. **Instruction 12:** Triangle Projection: Map the LUT result to the physical bit-offset, resolving the Atomic Polytope within the Row manifold.
13. **Instruction 13:** Commitment/Return: Write the new occupancy to the 15-bit metadata and return the start address S.

Appendix D: Time Trials

To verify the "No-Search" claim of the iscandar mechanism, a parallel Shadow Map (SM) is utilized during laboratory trials. This shadow map is a 3D integer array of identical dimensions to the bit-lattice ($32 \times 32 \times 32$). Every time a sector is accessed—whether to set an occupancy bit or clear it—the corresponding shadow coordinate is incremented.

```
∇ Coord ← ALLOC_SHADOWED
[1] S3 ← (32 | SEED_P)          % K1: Resolve Tier-3 Plane index
[2] S2 ← (32 | SEED_R)          % K2: Resolve Tier-2 Row index
[3] S1 ← 1ι ~Lattice[S3;S2;]    % K3: Resolve Tier-1 Bit index
[4] Coord ← (S3, S2, S1)        % K4: Assemble 3D Coordinate
[5] Lattice[S3;S2;S1] ← 1       % K5: Occupy sector in Lattice
[6] SM[S3;S2;S1] +← 1           % ** TOUCH: Increment shadow counter
[7] T2[S3;S2] -← 1              % K6: Update Tier-2 availability
[8] T3[S3] -← 1                 % K7: Update Tier-3 availability
∇

∇ FREE_SHADOWED Coord
[1] (S3 S2 S1) ← Coord          % K8: Decompose Coordinate
[2] Lattice[S3;S2;S1] ← 0       % K9: Release sector in Lattice
[3] SM[S3;S2;S1] +← 1           % ** TOUCH: Increment shadow counter
[4] T2[S3;S2] +← 1              % K10: Increment Tier-2 availability
[5] T3[S3] +← 1                 % K11: Increment Tier-3 availability
[6] SEED_P ← (32 | S3 + 1)      % K12: Advance T3 seed
[7] SEED_R ← (32 | S2 + 1)      % K13: Advance T2 seed
∇
```

Verification Measurements

The shadow map provides the following forensic data points:

- **Total Displacement:** The sum of all values in SM equals the total count of completed requests.
- **Zero Hunting:** No coordinate in the shadow map is touched unless it is the direct target of a resolution or a release.
- **Temporal Invariance:** The "work" required for any memory request is exactly one unit of physical displacement, confirming $CV = 0$.

8 Uranus 5 Verification: Source and Performance Report

The following section presents the raw APL test harness used within the Uranus 5 environment alongside the resulting architectural equivalence report. This documentation serves as the empirical verification of the $O(1)$ constant-path architecture.

8.1 Uranus 5 Source Code (APL)

```
=== PROJECT ISCANDAR-OZ: URANUS 5 INSTRUCTIONAL SUITE ===
```

```
 Geometry: 32-Radix (32,768 Sectors) | Determinism: Verified
 Metric: Discrete Ops / Six-Sigma Reliability
IO←0  ML←1

 RESET
  Lattice←32 32 321  T2←32 321  T3←321
  IC←0 0 0 0  [Logical, Search, Memory, Flow]
  Cycles←1000  Log←Cycles0


 R←ALLOC;P;Row;B
  P←T31  Row←(T2[P;])1  B←(Lattice[P;Row;])1  IC[1]+←3
  Lattice[P;Row;B]←0  IC[2]+←1
  T2[P;Row]←/Lattice[P;Row;]  T3[P]←/T2[P;]  IC[0]+←2  IC[2]+←2
  R←P Row B  IC[3]+←1


 FREE Addr;P;Row;B
  (P Row B)←Addr  IC[3]+←1
  Lattice[P;Row;B]←1  T2[P;Row]←1  T3[P]←1  IC[2]+←3


 --- LABORATORY EXECUTION ---
RESET
:For i :In Cycles
    Start_Count ← +/IC
    FREE ALLOC
    Log[i] ← (+/IC) - Start_Count
:EndFor

 --- Six-Sigma Statistics (Instructional Flux) ---
Mean_Ops ← (+/Log) ÷ Cycles
Std_Dev ← (((+/(Log - Mean_Ops)*2) ÷ Cycles) * 0.5)
CV ← Std_Dev ÷ (Mean_Ops + 1E¯10)
```

8.2 Uranus 5 Performance Report

```
'-----------------------------------------------'
' ISCANDAR-OZ: ARCHITECTURAL EQUIVALENCE REPORT '
'-----------------------------------------------'
'GOLD STANDARD (Theoretical K) | 13'
'URANUS 5 (Observed Variance)  | 0'
'COEFF OF VARIATION            | 0'
'--------------------|-------------|-------------'
'CLASS               |  RAW COUNT  |  PER CYCLE  '
'--------------------|-------------|-------------'
'Logical Intensity   | 2000        | 2'
'Search Complexity   | 3000        | 3'
```

```
'Memory Pressure     | 6000        | 6'
'Flow Overhead       | 2000        | 2'
'--------------------|-------------|-------------'
'EQUIVALENCE STATUS  | VERIFIED'
'DETERMINISM         | O(1) CONSTANT PATH CONFIRMED'
'-------------------------------------------------'
```

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